Applied Time Series Analysis
STA457 Lecture Notes

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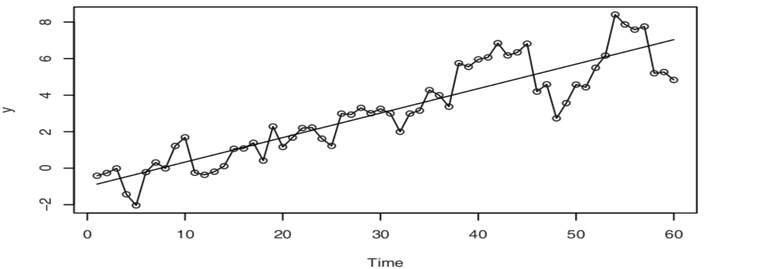
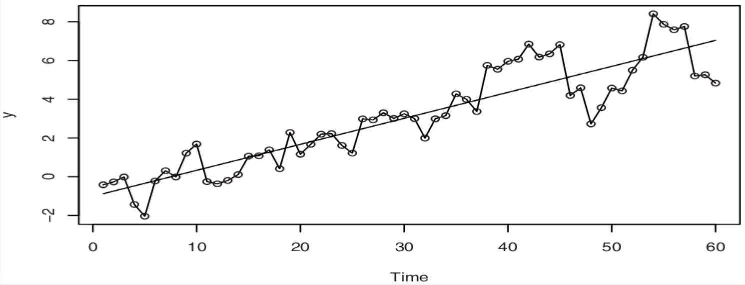
Trends and Model Fitting

Motivating Example



Motivating Example



	Estimate	Std. Error	t-value	$Pr(>\|t\|)$
Intercept	-1.007888	0.297245	-3.39	0.00126
Time	0.134087	0.008475	15.82	< 0.0001

Residual standard error	1.137	with 58 degrees of freedom
Multiple R-Squared	0.812	
Adjusted R-squared	0.809	
F-statistic	250.3	with 1 and 58 df; p-value < 0.0001

Trends

1. Time series exhibiting *trend* over time have a mean function that is some simple function (not necessarily constant) of time.

2. The example random walk graph from Chapter 2 showed an upward trend, but we know that a random walk process has constant mean zero.

3. That upward trend was simply a characteristic of that one particular random realization of the random walk.

4. If we generated other realizations of the random walk process, they would exhibit different trends.

5. Such trends could be called *stochastic trends* since they are just random and not fundamental to the underlying process.

6. The most important type of trend is a *deterministic trend*, which is related to the real nature of the process.

7. Often a time series process consists of some specified trend, plus a random component.

8. We commonly express such time series models using the form $Y_t = \mu_t + X_t$, where μ_t is a trend and X_t is a random process with mean zero for all t.

Regression Methods

1. Now, we consider several common nonconstant mean trend models: linear, quadratic, seasonal means and cosine trends.
2. A linear trend is expressed as:
$$\mu_t = \beta_0 + \beta_1 t \tag{1}$$
3. The least squares method chooses the estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize the least squares criterion:
$$Q(\beta_0, \beta_1) = \sum_{t=1}^{N} [y_t - (\beta_0 + \beta_1 t)]^2 \tag{2}$$
4. A quadratic trend is expressed as:
$$\mu_t = \beta_0 + \beta_1 t + \beta_2 t^2 \tag{3}$$
5. The least squares method minimizes the least squares criterion:
$$Q(\beta_0, \beta_1, \beta_2) = \sum_{t=1}^{N} [y_t - (\beta_0 + \beta_1 t + \beta_2 t^2)]^2 \tag{4}$$
6. Before fitting a linear or quadratic model, it is important to ensure that this trend truly represents the deterministic nature of the time series process and is not simply an artifact of the randomness of that realization of the process as in the random walk example.

Precision of Sample Mean

1. In most statistics texts, the first inferential problem considered is the estimation of a population mean using a sample mean. In time series analysis, however, the situation is rather different.

2. The sample mean should only be considered as a summary statistic for time series data that are thought to have come from a stationary process. Even when this is so, it is important to realize that the statistical properties of the sample mean are quite different from those that usually apply.

3. Suppose we have a time series data $\{x_t\}_{t=1}^{N}$ realized from some stationary stochastic process having mean μ, variance σ^2 and theoretical ACF ρ_k. For the sample mean $\bar{X} = \frac{1}{N}\sum_{t=1}^{N} X_t$, the usual result for independent observations is that

$$Var[\bar{X}] = \frac{\sigma^2}{N}. \tag{5}$$

4. For correlated observations (the case of time series data), we have

$$Var[\bar{X}] = \frac{\sigma^2}{N}\left[1 + 2\sum_{k=1}^{N}(1 - \frac{k}{N})\rho_k\right] \tag{6}$$

   and this quantity can differ considerably from σ^2/N when autocorrelations are significant. If the series $\{x_t\}$ is just a realization of some white noise process, then $\rho_k = 0$ for $k > 0$ and $Var[\bar{X}]$ reduces to simply σ_w^2/N.

Exercise 2.17.

Let $\{X_t\}$ be stationary with autocovariance function γ_h, and let $\bar{X} = \frac{1}{n}\sum_{t=1}^{n} X_t$. Show that

$$Var[\bar{X}] = \frac{\gamma_0}{n} + \frac{2}{n}\sum_{h=1}^{n-1}(1-\frac{h}{n})\gamma_h = \frac{1}{n}\sum_{h=-n+1}^{n-1}(1-\frac{|h|}{n})\gamma_h \tag{7}$$

Solution.

$$Var[\bar{X}] = \frac{1}{n^2}Var[\sum_{t=1}^{n}X_t] = \frac{1}{n^2}Cov(\sum_{t=1}^{n}X_t, \sum_{s=1}^{n}X_s) = \frac{1}{n^2}\sum_{t=1}^{n}\sum_{s=1}^{n}\gamma_{t-s} \tag{8}$$

Now, make the change of variable $t-s=h$ and $t=j$ in the double summation. The range of the summation $\{1 \le t \le n, 1 \le s \le n\}$ is transformed into $\{1 \le j \le n, 1 \le j-h \le n\} = \{h+1 \le j \le n+h, 1 \le j \le n\}$ which may be written as $\{h > 0, h+1 \le j \le n\} \cup \{h \le 0, 1 \le j \le n+h\}$. Thus,

$$\begin{aligned}
Var[\bar{X}] &= \frac{1}{n^2}\left[\sum_{h=1}^{n-1}\sum_{j=h+1}^{n}\gamma_h + \sum_{h=-n+1}^{0}\sum_{j=1}^{n+h}\gamma_h\right] \\
&= \frac{1}{n^2}\left[\sum_{h=1}^{n-1}(n-h)\gamma_h + \sum_{h=-n+1}^{0}(n+h)\gamma_h\right] \\
&= \frac{1}{n}\sum_{h=-n+1}^{n-1}(1-\frac{|h|}{n})\gamma_h
\end{aligned}$$

In order to get the first expression in the exercise, you can use the fact that $\gamma_h = \gamma_{-h}$.

Exercise 2.18.

Let $\{X_t\}$ be stationary with autocovariance function γ_h, and let $S^2 = \frac{1}{n-1}\sum_{t=1}^{n}(X_t - \bar{X})^2$ be the sample variance. Find $\mathbb{E}[S^2]$.

Solution.

$$\begin{aligned}
\sum_{t=1}^{n}(X_t-\mu)^2 &= \sum_{t=1}^{n}(X_t-\bar{X}+\bar{X}-\mu)^2\\
&= \sum_{t=1}^{n}(X_t-\bar{X})^2+\sum_{t=1}^{n}(\bar{X}-\mu)^2+2\sum_{t=1}^{n}(X_t-\bar{X})(\bar{X}-\mu)\\
&= \sum_{t=1}^{n}(X_t-\bar{X})^2+n(\bar{X}-\mu)^2+2(\bar{X}-\mu)\sum_{t=1}^{n}(X_t-\bar{X})=\sum_{t=1}^{n}(X_t-\bar{X})^2+n(\bar{X}-\mu)^2
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}[S^2] = \mathbb{E}\left[\frac{1}{n-1}\sum_{t=1}^{n}(X_t-\bar{X})^2\right] &= \frac{1}{n-1}\mathbb{E}\left[\sum_{t=1}^{n}(X_t-\mu)^2-n(\bar{X}-\mu)^2\right]\\
&= \frac{1}{n-1}\left[\sum_{t=1}^{n}\mathbb{E}[(X_t-\mu)^2]-n\mathbb{E}[(\bar{X}-\mu)^2]\right]=\frac{1}{n-1}[n\gamma_0-nVar[\bar{X}]]\\
&= \frac{1}{n-1}\left[n\gamma_0-n\{\frac{\gamma_0}{n}+\frac{2}{n}\sum_{h=1}^{n-1}(1-\frac{h}{n})\gamma_h\}\right]=\gamma_0-\frac{2}{n-1}\sum_{h=1}^{n-1}(1-\frac{h}{n})\gamma_h
\end{aligned}$$

Precision of Sample Mean

1. As an example, consider the moving average process $X_t = W_t - \frac{1}{2}W_{t-1}$ where $\{W_t\}$ is a white noise process. Then,

$$\rho_k = \begin{cases} -0.4 & \text{if } k = 1 \\ 0 & \text{if } k > 1 \end{cases} \tag{12}$$

In this case, we have

$$Var[\bar{X}] = \frac{\gamma_0}{N}\left[1 + 2(1 - \frac{1}{N}(-0.4))\right] = \frac{\gamma_0}{N}\left[1 - 0.8(\frac{N-1}{N})\right] \overset{N\to\infty}{\approx} 0.2\frac{\gamma_0}{N} \tag{13}$$

2. We see that the negative autocorrelation at lag 1 has improved the estimation of the mean as compared to the case where the sample mean of a white noise (purely random) process is estimated. **Because the series tends to oscillate back and forth across the mean (why?), the estimate of the sample mean is more precise**.

3. On the other hand, if $\rho_k \geq 0$ for all $k \geq 1$, $Var[\bar{X}]$ will be larger than $\frac{\gamma_0}{N}$. Here the positive autocorrelations make estimation of the sample mean more difficult than in the white noise case.

4. In general, some autocorrelations will be positive and some negative, and Equation (6) must be used to assess the total effect.

Precision of Sample Mean

1. For many stationary processes, the autocorrelation function decays quickly enough with increasing lags that
$$\sum_{k=0}^{\infty} |\rho_k| < \infty \tag{14}$$
Under this assumption and given a large sample size N, the following useful approximation follows:
$$Var[\bar{X}] \approx \frac{\gamma_0}{N}\left[\sum_{k=-\infty}^{\infty} \rho_k\right] \tag{15}$$
Notice that in this approximation the variance is inversely proportional to the sample size N.

2. As an example, we can consider AR(1) whose autocorrelation function is $\rho_k = \phi^{|k|}$ where $|\phi| < 1$. Then, by summing the geometric series we obtain
$$Var[\bar{X}] = \frac{1+\phi}{1-\phi}\frac{\gamma_0}{N}. \tag{16}$$
This means that the equivalent number of independent observations is $N\frac{1-\phi}{1+\phi}$. When $\phi > 0$, the positive autocorrelation means that there is less information in the data than might be expected in regard to estimating the mean. On the other hand, when $\phi < 0$, there is more information.

Precision of Sample Mean

1. For a nonstationary process (but with a constant mean), the precision of the sample mean as an estimate of μ can be strikingly different.

2. As an example, consider a random walk process. Then, we obtain

$$
\begin{aligned}
Var[\bar{X}] &= \frac{1}{N^2} Var\left[\sum_{i=1}^{N}\sum_{j=1}^{i} W_j\right] \\
&= \frac{1}{N^2} Var[W_1 + 2W_2 + 3W_3 + \cdots + NW_N] \\
&= \frac{\sigma_w^2}{N^2}\sum_{k=1}^{N} k^2 \\
&= \sigma_w^2 (2N+1)\frac{(N+1)}{6N}
\end{aligned}
$$

3. Notice that in this special case the variance of our estimate of the mean actually increases as the sample size N increases. Clearly this is unacceptable, and we need to consider other estimation techniques for nonstationary time series.

Cyclical or Seasonal Trends

1. The *seasonal means* approach represents the mean function with a different parameter for each level.

2. For example, suppose each measure time is a different month, and we have observed data over a period of several years.

3. The seasonal means model might then specify a different mean response for each of the 12 months:

$$\mu_t = \begin{cases} \beta_1 & \text{for } t = 1, 13, 25, \ldots \\ \beta_2 & \text{for } t = 2, 14, 26, \ldots \\ \vdots & \\ \beta_{12} & \text{for } t = 12, 24, 36, \ldots \end{cases} \tag{18}$$

4. This is similar to an ANOVA model in which the parameters are the mean response values for each factor level.

5. This model, however, makes no assumption about the shape of the mean response function over time.

Harmonic Regression for Cosine Trends

1. A more specific model for seasonal case might assume that the mean response varies over time in some regular manner.

2. For example, a model can assume that the mean response across time follows a periodic pattern such as:
$$\mu_t = \beta \cos(2\pi f t + \phi) \tag{19}$$
where β is the amplitude, f the frequency and ϕ the phase.

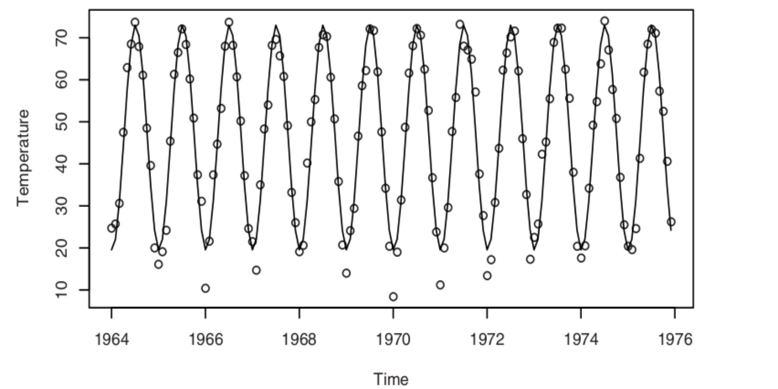
3. To fit the model, it is useful to consider a transformation of the mean response function:
$$\mu_t = \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t) \tag{20}$$
where $\beta = \sqrt{\beta_1^2 + \beta_2^2}$ and $\phi = \arctan(-\beta_2/\beta_1)$.

4. Then, β_1 and β_2 enter the regression equation linearly and we can estimate them using linear regression with $\cos(2\pi f t)$ and $\sin(2\pi f t)$ as predictors, along with an intercept term:
$$\mu_t = \beta_0 + \beta_1 \cos(2\pi f t) + \beta_2 \sin(2\pi f t) \tag{21}$$

5. Note that the amplitude β is the height of the cosine curve from its midpoint to its top, and the frequency f is the reciprocal of the period, which measures how often the curve's pattern repeat itself.

Harmonic Regression for Cosine Trends

1. In any practical example, we must be careful how we measure time, as our choice of time measurement will affect the values of the frequencies of interest. For example, if we have monthly data but use $1, 2, 3, \ldots$ as our time scale, then $1/12$ would be the most interesting frequency, with a corresponding period of 12 months.



Reliability and Efficiency of Regression Estimates

1. We assume that the series is represented as $Y_t = \mu_t + X_t$, where μ_t is a deterministic trend of the kind considered in seasonal model and $\{X_t\}$ is a zero-mean stationary process with autocovariance and autocorrelation functions γ_k and ρ_k, respectively.

2. Ordinary regression estimates parameters in a linear model according to the criterion of least squares regardless of whether we are fitting linear time trends, seasonal means, cosine curves, or whatever.

3. We first consider the easiest case, the seasonal means. Basically, the least squares estimates of the seasonal means are just seasonal averages; thus, if we have N years of monthly data, we can write the estimate for the mean for the jth season as

$$\hat{\beta}_j = \frac{1}{N} \sum_{i=0}^{N-1} Y_{j+12i} \tag{22}$$

4. Since $\hat{\beta}_j$ is an average like $\bar{Y}$ but uses only uses every 12th observation, we can write

$$Var[\hat{\beta}_j] = \frac{\gamma_0}{N} \left[1 + 2 \sum_{k=1}^{N-1} (1 - \frac{k}{N} \rho_{12k}) \right] \quad \text{for } j = 1, 2, \ldots, 12 \tag{23}$$

5. Note that only the seasonal autocorrelations, $\rho_{12}, \rho_{24}, \rho_{36}, \ldots$ enter into equation presented above.

Reliability and Efficiency of Regression Estimates

1. Now, consider the cosine trends and their associated models. For any frequency of the form $f = \frac{m}{n}$, where m is an integer satisfying $1 \leq m < \frac{n}{2}$, explicit expressions are available for the estimates $\hat{\beta}_1$ and $\hat{\beta}_2$ (the amplitudes of the cosine and sine):

$$\hat{\beta}_1 = \frac{2}{n} \sum_{t=1}^{n} \left[\cos(\frac{2\pi mt}{n}) Y_t \right] \tag{24}$$

$$\hat{\beta}_2 = \frac{2}{n} \sum_{t=1}^{n} \left[\sin(\frac{2\pi mt}{n}) Y_t \right] \tag{25}$$

2. These are effectively the correlations between the time series $\{Y_t\}$ and the cosine and sine waves with frequency m/n.

3. Because these are linear functions of $\{Y_t\}$, we may evaluate their variances using

$$Var[\hat{\beta}_1] = \frac{2\gamma_0}{n} \left[1 + \frac{4}{n} \sum_{s=2}^{n} \sum_{t=1}^{s-1} \cos(\frac{2\pi mt}{n}) \cos(\frac{2\pi ms}{n}) \rho_{s-t} \right] \tag{26}$$

4. If $\{X_t\}$ is white noise, then we get just $\frac{2\gamma_0}{n}$. However, if $\rho_1 \neq 0$ and $\rho_k = 0$ for $k > 1$, and $m/n = 1/12$, then the variance reduces to

$$Var[\hat{\beta}_1] = \frac{2\gamma_0}{n} \left[1 + \frac{4\rho_1}{n} \sum_{t=1}^{n-1} \cos(\frac{\pi t}{6}) \cos(\frac{\pi t + 1}{6}) \right] \tag{27}$$

Reliability and Efficiency of Regression Estimates

1. To illustrate the effect of the cosine terms, let's compute some representative values:

n	$Var(\hat{\beta}_1)$
25	$\left(\frac{2\gamma_0}{n}\right)(1 + 1.71\rho_1)$
50	$\left(\frac{2\gamma_0}{n}\right)(1 + 1.75\rho_1)$
500	$\left(\frac{2\gamma_0}{n}\right)(1 + 1.73\rho_1)$
∞	$\left(\frac{2\gamma_0}{n}\right)\left(1 + 2\rho_1 \cos\left(\frac{\pi}{6}\right)\right) = \left(\frac{2\gamma_0}{n}\right)(1 + 1.732\rho_1)$

2. If $\rho_1 = -0.4$, then the large sample multiplier in the last equation is

$$1 + 1.732(-0.4) = 0.307$$

meaning that the variance is reduced by about 70% when compared with the white noise case.

Seasonal Means vs. Cosine Trends

1. In some circumstances, seasonal means and cosine trends could be considered as competing models for a cyclical trend.

2. If the simple cosine model is an adequate model, then the question is ”How much do we lose information if we use the less parsimonious seasonal means model?” To answer this question, we can compare the estimates of the trend at comparable time points.

3. For instance, consider the two estimates for the trend in January, i.e., μ_1. With seasonal means, this estimate is just the January average, which has variance given by

$$Var[\hat{\beta}_1] = \frac{\gamma_0}{N}\left[1 + 2\sum_{k=1}^{N-1}(1 - \frac{k}{N}\rho_{12k})\right] \tag{28}$$

4. With the cosine trend model, the corresponding estimate is

$$\hat{\mu}_1 = \hat{\beta}_0 + \hat{\beta}_1 \cos(\frac{2\pi}{12}) + \hat{\beta}_2 \sin(\frac{2\pi}{12}) \tag{29}$$

5. To compute the variance of this estimate, we need the fact that with this model, the estimates $\hat{\beta}_0$, $\hat{\beta}_1$ and $\hat{\beta}_2$ are uncorrelated. For the cosine model, we then have

$$Var[\hat{\mu}_1] = Var[\hat{\beta}_0] + Var[\hat{\beta}_1]\left[\cos(\frac{2\pi}{12})\right]^2 + Var[\hat{\beta}_2]\left[\sin(\frac{2\pi}{12})\right]^2 \tag{30}$$

Seasonal Means vs. Cosine Trends

1. For our comparison, let us assume that the stochastic component is white noise. Then, the variance of our estimate in the seasonal means model is just $\frac{\gamma_0}{N}$.

2. For the cosine model, we obtain
$$Var[\hat{\mu}_1] = \frac{\gamma_0}{n}\left\{1 + 2\left[\cos(\frac{\pi}{6})\right]^2 + 2\left[\sin(\frac{\pi}{6})\right]^2\right\} = 3\frac{\gamma_0}{n} \tag{31}$$

3. Thus, the ratio of the standard deviation in the cosine model to that in the seasonal means model is
$$\sqrt{\frac{\frac{3\gamma_0}{n}}{\frac{\gamma_0}{N}}} = \sqrt{\frac{3N}{n}}.$$

4. For instance, if we consider $n = 144$ and $N = 12$, we get
$$\sqrt{\frac{3(12)}{144}} = 0.5 \tag{32}$$
which implies that, in the cosine model, we estimate the January effect with a standard deviation that is only half as large as it would be if we estimated with a seasonal means model. This is a substantial gain!

5. Note that this assumes that the cosine trend plus white noise model is the correct model.

Properties of the Least Squares Estimators

1. Section 3.4 gives some theoretical results about the variance of the least squares estimators.

2. When the random components $\{X_t\}$ is not white noise, the least squares estimators are not *best linear unbiased estimators* (BLUEs).

3. However, under some general conditions on the random component $\{X_t\}$, it is known that if the trend over time is either

    1. polynomial
    2. a trigonometric polynomial
    3. seasonal means
    4. any linear combination of these

    then the least squares estimates of the trend coefficients have approximately the same variance as the BLUEs, for large sample sizes.

Interpreting the Regression Output

1. The standard regression routines (like in R), calculate least squares estimates of the unknown regression coefficients (the βs). The same routines also give us the *residual standard deviation*, which estimates $\sqrt{\gamma_0}$, the standard deviation of $\{X_t\}$:

$$s = \sqrt{\frac{1}{n-p}\sum_{t=1}^{N}(y_t - \hat{\mu}_t)^2} \tag{33}$$

where p is the number of parameters estimated in $\hat{\mu}_t$ and $n-p$ is the so-called *degrees of freedom* for s.

2. The value of s gives an absolute measure of the goodness of fit of the estimated trend. The smaller the value of s, the better the fit. The value of s can be used to compare alternative trend models.Note that s possesses a unit which makes its interpretation difficult.

3. A unitless measure of the goodness of fit of the trend is the value of R^2, also called the *coefficient of determination*. One interpretation of R^2 is that it is the square of the sample correlation coefficient between the observed series and the estimated trend. It is also the fraction of the variation in the series that is explained by the estimated trend. Any *adjusted* R^2 measure is similar, but penalizes models with more parameters. Like s, the adjusted R^2 can be used to compare different trend models; a higher adjusted R^2 indicates a better model.

Model Diagnostics

Residual Analysis

1. The unobserved stochastic component $\{X_t\}$ can be estimated or predicted, by the *residual*

$$\hat{x}_t = y_t - \hat{\mu}_t \tag{34}$$

2. Note that we use the term estimate for the guess of an unknown parameter, and the term predictor for an estimate of an unobserved random variable.

3. We call $\hat{x}_t$ the residual corresponding to the tth observation.

4. If the trend model is reasonably correct, then the residuals should behave roughly like a true stochastic component, and various assumptions about the stochastic component can be assessed by looking at the residuals.

5. If the stochastic component is white noise, then the residuals should behave like independent (normal) random variables with zero mean and standard deviation s.

6. We can plot the residuals against time and look for any patterns that might deviate from white noise.

7. We can also plot the residuals against the fitted trend values $\hat{\mu}_t$ and look for patterns. For example, does the variation in residuals change as the fitted trend values change? If we see notable patterns in the plots, we may rethink whether our model assumptions are appropriate.

The use of Correlograms

1. Suppose $x_1, x_2, \ldots, x_N$ are observations on independent and identically distributed random variables with arbitrary mean. Then, it can be shown that

$$\mathbb{E}[r_k] \approx -\frac{1}{N} \tag{35}$$

$$Var[r_k] \approx \frac{1}{N} \tag{36}$$

and that r_k is asymptotically normally distributed. Thus, having plotted the correlogram, we can check for randomness by plotting approximate 95% confidence limits at

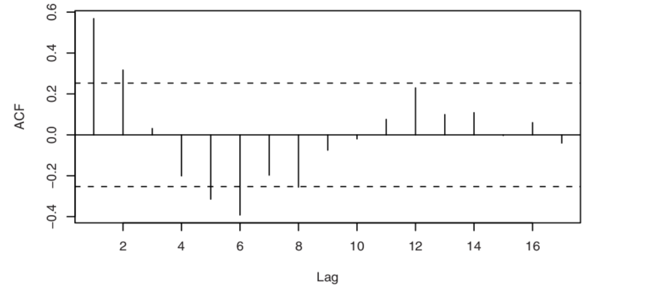
$$-\frac{1}{N} \pm \frac{1.96}{\sqrt{N}} = \frac{-1 \pm 1.96\sqrt{N}}{N} \approx \frac{\pm 1.96}{N} = \pm\frac{1.96}{N}. \tag{37}$$

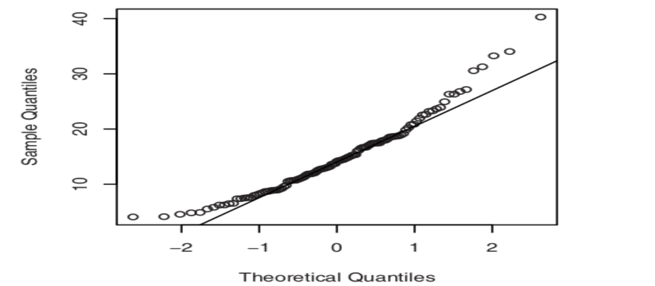
2. A correlogram where the values of r_k do not come down to zero reasonably quickly, indicates nonstationarity and so the series needs to be differenced.

3. The ACF of an MA(q) process is easy to recognize as it cuts off at lag q.

4. The ACF of an AR(p) process is rather more difficult to categorize, as it is a mixture of damped exponential and sinusoids and dies out slowly.

Outputs for Straight Line Fit of Random Walk





Outputs for Temperature Seasonal Means

Exhibit 3.8 **Residuals versus Time for Temperature Seasonal Means**

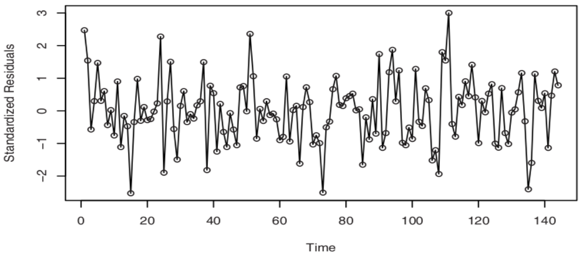
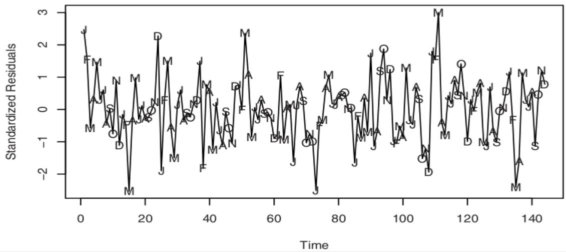


Exhibit 3.9 **Residuals versus Time with Seasonal Plotting Symbols**



Outputs for Temperature Seasonal Means

Exhibit 3.10 Standardized Residuals versus Fitted Values for the Temperature Seasonal Means Model

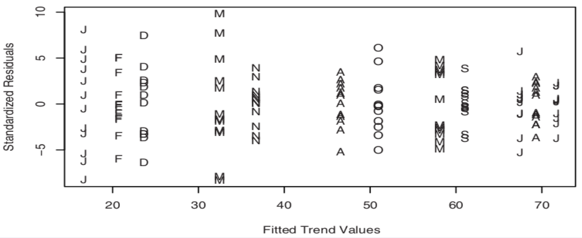
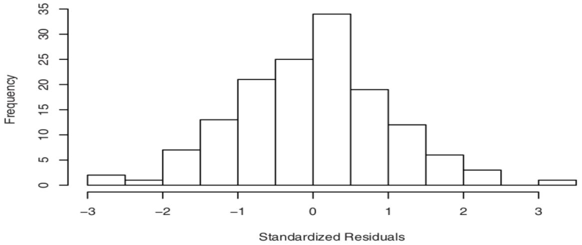


Exhibit 3.11 Histogram of Standardized Residuals from Seasonal Means Model



Outputs for Temperature Seasonal Means

Exhibit 3.12 Q-Q Plot: Standardized Residuals of Seasonal Means Model

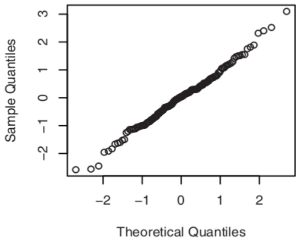
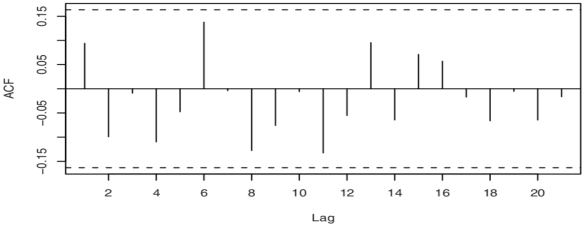
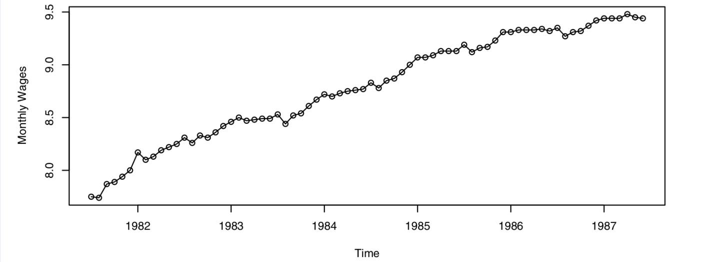


Exhibit 3.13 Sample Autocorrelation of Residuals of Seasonal Means Model



Exercise 3.5.

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> wages.lm=lm(wages~time(wages)); summary(wages.lm); y=rstudent(wages.lm)
```

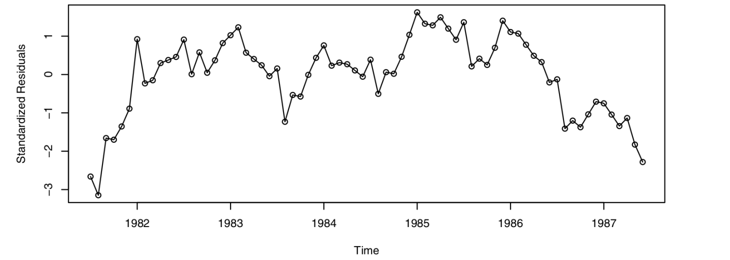
```
Call:
lm(formula = wages ~ time(wages))

Residuals:
     Min       1Q   Median       3Q      Max 
-0.23828 -0.04981  0.01942  0.05845  0.13136 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept) -5.490e+02  1.115e+01  -49.24   <2e-16 ***
time(wages)  2.811e-01  5.618e-03   50.03   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08257 on 70 degrees of freedom
Multiple R-Squared: 0.9728,     Adjusted R-squared: 0.9724 
F-statistic:  2503 on 1 and 70 DF,  p-value: < 2.2e-16
```

Exercise 3.5.




```
> wages.lm2=lm(wages~time(wages)+I(time(wages)^2))
> summary(wages.lm2); y=rstudent(wages.lm)
```

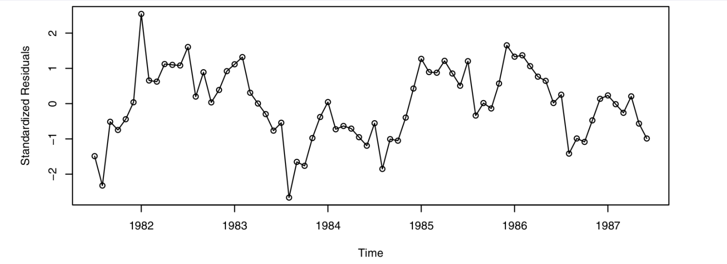
```
Call:
lm(formula = wages ~ time(wages) + I(time(wages)^2))

Residuals:
      Min        1Q    Median        3Q       Max
-0.148318 -0.041440  0.001563  0.050089  0.139839

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)      -8.495e+04  1.019e+04  -8.336 4.87e-12 ***
time(wages)       8.534e+01  1.027e+01   8.309 5.44e-12 ***
I(time(wages)^2) -2.143e-02  2.588e-03  -8.282 6.10e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.05889 on 69 degrees of freedom
Multiple R-Squared: 0.9864,     Adjusted R-squared: 0.986
F-statistic:  2494 on 2 and 69 DF,  p-value: < 2.2e-16
```

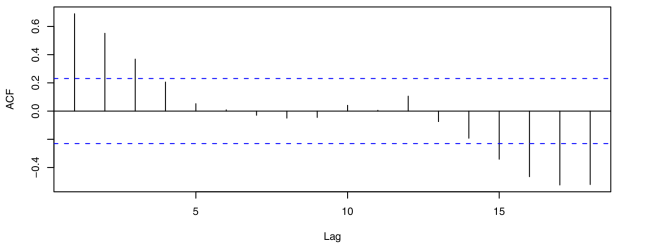
Exercise 3.5.

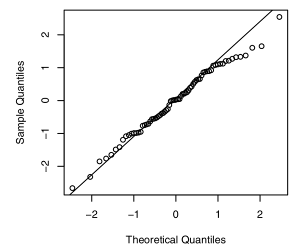


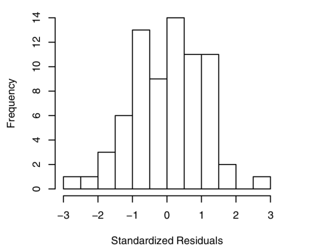

```
> runs(rstudent(wages.lm2))
```

```
$pvalue
[1] 1.56e-07
$observed.runs
[1] 15
$expected.runs
[1] 36.75
$n1
[1] 33
$n2
[1] 39
$k
[1] 0
```

Exercise 3.5.








```
Shapiro-Wilk normality test
data:  rstudent(wages.lm2)
W = 0.9887, p-value = 0.7693
```